MoCHII Atomic-Data Fitting: CHIANTI Cooling and n-level Fits

Abstract

MoCHII evaluates all metal-line atomic data from analytic fits; the raw CHIANTI database is never parsed at runtime. The Python fitting scripts under `tools/fitting/` are the auditable source of every coefficient in `data/atomic/`. This memo documents the fitting procedure and its two products: the cooling-function fits that carry the thermal-loop line cooling in the $n_e \to 0$ limit, 28 ions in the historical CHIANTI fit set (H I and 27 metal ions) — suppressed to the local electron density at run time by a (T, n_e) table built from the n-level solve — and the n-level data files (level energies, A-values, and Chebyshev fits of the effective collision strengths in Burgess–Tully scaled space) that supply the output-time line emissivities. The historical CHIANTI fit set was checked against the CHIANTI v11.0.2 source data and against PyNeb, and the cooling fits are additionally checked channel by channel against an independent photoionization code on a fixed gas state (`docs/MoCHII_cooling_analysis.pdf`). The fitted files reproduce direct CHIANTI evaluation to $\lesssim 0.5\%$ in the classic diagnostic ratios; the remaining differences from PyNeb (up to 6–10% for the density diagnostics near their critical densities) trace to PyNeb's independent choice of collision strengths and A-values, not to fit error.

Current selection and scope (2026-10-11). The production registry has 43 metal stages and 32 transitions; the 77-stage, 66-transition expansion is staged but not installed. The default `metal_excitation_model = 'mocassin_v2'` reads metal cooling and n-level files from `data/atomic/excitation/mocassin_v2/`; the `'extended'` selection reads `data/atomic/excitation/extended/`. The selections also determine emitted lines and charge-transfer product populations. Hydrogen excitation remains under `data/atomic/`. The two selections currently contain 55 atoms/571 levels/49 fits/763 terms and 62 atoms/1951 levels/38 fits/377 terms, respectively. These file counts do not establish atomic accuracy. Ca VII remains unaccepted: 27 levels, 310 rates, and four negative-source gaps remain unresolved. The Mg I constant model has been numerically checked over 1000–100000 K, but its atomic accuracy has not been verified. Tables and figures below describe the historical CHIANTI fitting set and have not been recomputed for the selected models.

1. Overview: two products, one design rule

The metal-line design requires that adding an ion be a data operation, not a code operation. The fitting scripts therefore produce, for each registered ion, self-contained coefficient files with provenance headers (CHIANTI version, population model, fit range, fit form, maximum fit error), of two kinds:

- **Cooling fits (thermal balance).** A smooth closed-form cooling function $\Lambda_{\rm ion}(T)$, cheap and smooth inside the thermal-balance bisection on $\log T$, in the $n_e \to 0$ limit of statistical equilibrium. This is the low-density *carrier*: at run time the thermal loop suppresses it to the local electron density with a (T, n_e) table built from the same n-level solve (`par%cooling_model = local_ne`, the default), so the thermal balance and the diagnostic line list are one physics.
- **n-level data (diagnostics).** Level energies, statistical weights, radiative A-values, and fitted effective collision strengths $\Upsilon(T)$ feeding the generic n-level solver (`nlevel_mod`), evaluated on the converged state for the line

output and, at setup, to build the (T, n_e) suppression table of the thermal balance (`nlevel_cooling_mod`). These supply the density-dependent emissivities and the diagnostic line ratios that the cooling fits cannot give.

The fitting scripts are seeded from EXHALE: `chianti_cooling.py` (copied from `EXHALE/cooling_data/`) reads the raw CHIANTI ASCII ion files (`.elvlc`, `.wgfa`, `.scups`) directly, with no ChiantiPy dependency, and applies the Burgess–Tully (1992) descaling of the stored scaled collision strengths. CHIANTI v11.0.2 lives at `~/RT_Codes/CHIANTI/dbase`.

2. The cooling function

For a transition $l \to u$ with energy gap ΔE_{lu}, the Maxwellian electron-impact excitation rate coefficient is

$$q_{lu}(T) \;=\; \frac{8.629 \times 10^{-6}}{g_l \sqrt{T}}\, \Upsilon_{lu}(T)\, e^{-\Delta E_{lu}/kT} \qquad [\mathrm{cm^3\,s^{-1}}], \tag{1}$$

with $\Upsilon_{lu}(T)$ the effective collision strength obtained by descaling the `.scups` data. In the optically thin, low-density limit every collisional excitation is followed by a radiative decay, so the line cooling normalized by $n_e\, n_{\rm ion}$ is

$$\Lambda_{\rm ion}(T) \;=\; \sum_{l \to u} f_l(T)\, q_{lu}(T)\, \Delta E_{lu} \qquad [\mathrm{erg\,cm^3\,s^{-1}}], \tag{2}$$

where $f_l(T)$ is the fractional population of the lower level. The fitting procedure evaluates the $n_e \to 0$ *limit of statistical equilibrium*: at $n_e \ll n_{\rm crit}$ of the ground-term fine structure the metastable levels are drained radiatively faster than collisions populate them, so every ion sits in its lowest level ($f_1 = 1$) and excitation happens out of that level only. Each excitation is followed by a radiative cascade that ultimately radiates the full ΔE_{1u} whatever route it takes down, so $\Lambda_{\rm ion}$ is the excitation power out of the ground level and is exactly what the n-level solve of Section 4. returns as $n_e \to 0$. ΔE_{lu} is the observed `.elvlc` level difference in both the Boltzmann factor of Eq. (1) and the radiated energy of Eq. (2); the `.scups` de is the Burgess–Tully *scaling* energy and enters only the scaled temperature. Detailed balance ties the excitation rate to the same level gap that appears in the equilibrium matrix, so the two energies must not be interchanged.

3. Cooling-function fits

Each $\Lambda_{\rm ion}(T)$ curve is fitted over 10^3–10^5 K with the multi-exponential form used throughout EXHALE,

$$\Lambda_{\rm ion}(T) \;=\; T^{-1/2} \sum_{i=1}^{N} A_i\, e^{-T_i/T}, \tag{3}$$

a Levenberg–Marquardt fit on $\log\Lambda$. In the $n_e \to 0$ limit Eq. (3) is the exact form of Eq. (2) apart from the slow temperature dependence of Υ: the T_i are the excitation temperatures $\Delta E_{1u}/k$ and the A_i their excitation-weighted strengths. The starting ladder is therefore built from the level energies themselves — the transitions out of the ground level are clustered in $\log(\Delta E_{1u}/k)$, weighted by the share of the cooling each carries across the fit range — and the fitter sweeps $N = 2 \ldots 8$ with six perturbed restarts at each N, stopping at the first N that holds the maximum error below 1%; where no N reaches it, the best fit found over all N is kept (`fit_cooling.py`). In this historical set, nine of the 28 ions end above 1%. Ions with only a few excitation blocks (the Cl ions have four transitions out of the ground term) are the hardest case, because a single exponential has a fixed amplitude and the temperature

dependence of Υ has to be carried by superposing neighboring terms; the three Cl ions end at 7–9%. Table 1 lists the fit quality; Figure 1 overlays the fits on the CHIANTI curves. `fit_cooling.py` writes `cooling_<el>_<stage>.txt` to its requested output directory (historically `data/atomic/`). Approved model installation places the matching runtime fits under `data/atomic/excitation/<model>/`.

Table 1: Cooling fits, 10^3–10^5 K, as recorded in the headers of `data/atomic/cooling_<el>_<stage>.txt` (generated 2026-07-25). The maximum relative error is against the direct CHIANTI evaluation of Eq. (2); "all T" is over the whole range, "$\Lambda > 10^{-3}\Lambda_{\max}$" only where the cooling is above that fraction of its maximum.

Ion	N	all T	$\Lambda > 10^{-3}\Lambda_{\max}$	Ion	N	all T	$\Lambda > 10^{-3}\Lambda_{\max}$
H I	4	0.52%	0.52%	S IV	3	0.61%	0.61%
C I	6	0.27%	0.27%	Ar III	2	3.90%	3.90%
C II	5	0.35%	0.35%	Ar IV	4	0.27%	0.27%
C III	6	5.48%	0.50%	Mg II	4	0.22%	0.12%
N I	5	0.58%	0.34%	Fe II	5	0.56%	0.56%
N II	5	0.66%	0.66%	Fe III	5	0.48%	0.48%
N III	5	0.70%	0.70%	Si II	6	0.91%	0.91%
O I	6	0.20%	0.20%	Si III	6	27.70%	11.29%
O II	4	0.27%	0.15%	Si IV	2	0.90%	0.90%
O III	5	0.32%	0.32%	Cl II	3	8.05%	8.05%
Ne II	5	0.40%	0.40%	Cl III	6	9.15%	9.15%
Ne III	5	0.34%	0.34%	Cl IV	2	7.30%	7.30%
S II	3	2.23%	2.16%	Ca II	4	1.03%	0.42%
S III	5	0.23%	0.23%	Ca V	2	4.34%	4.34%

A caveat carried in every file header: Eq. (3) is low-density-limit cooling. Transitions with low critical density (the fine-structure IR lines, most sharply [C II] 158 μm with $n_{\rm crit} \sim 50$ cm^{-3} and the [O III] 52, 88 μm pair; [O II] $\lambda\lambda$3726, 3729; [S II] $\lambda\lambda$6717, 6731) saturate above $n_{\rm crit}$, where the fit overestimates the cooling. The thermal loop therefore treats Eq. (3) as the $n_e \to 0$ carrier and suppresses it to the local density: on the Cloudy c25.00 state of the HII40 benchmark ($n_e \simeq 110$ cm^{-3}; the fixed-state comparison of `MoCHII_cooling_analysis.pdf`, measured 2026-07-25) the fitted line cooling exceeds the Cloudy inventory by 11.6%, of which +5.1 points come from [O III] and +3.9 from [C II], while the same ions evaluated with the n-level solve at the local n_e land within 2.4% and 6% and bring the summed line cooling to 1.01 times Cloudy's. MoCHII therefore builds a (T, n_e) suppression table from the same n-level solve at setup and applies it in the thermal loop by default (`par%cooling_model = local_ne`); `low_density` uses the fits directly. The emissivities themselves always come from the n-level data.

4. n-level data with fitted collision strengths

For each ion the lowest n levels are exported — enough for the optical/IR diagnostics (Table 2); extra levels are harmless to the solver. The file `data/atomic/nlevel_<el>_<stage>.txt` contains three blocks:

- `NLEV`: level index, statistical weight g, energy in cm^{-1} (observed values preferred, theoretical as fallback);

- `NRAD`: radiative rates A_{ul} from `.wgfa` (duplicate rows summed);
- `NUPS`: one line for each collisional transition — the Burgess–Tully type, scaling parameter C, transition energy in Ryd, the fitted domain $[s_{\rm lo}, s_{\rm hi}]$, a log flag, and the Chebyshev coefficients, followed by the recorded maximum fit

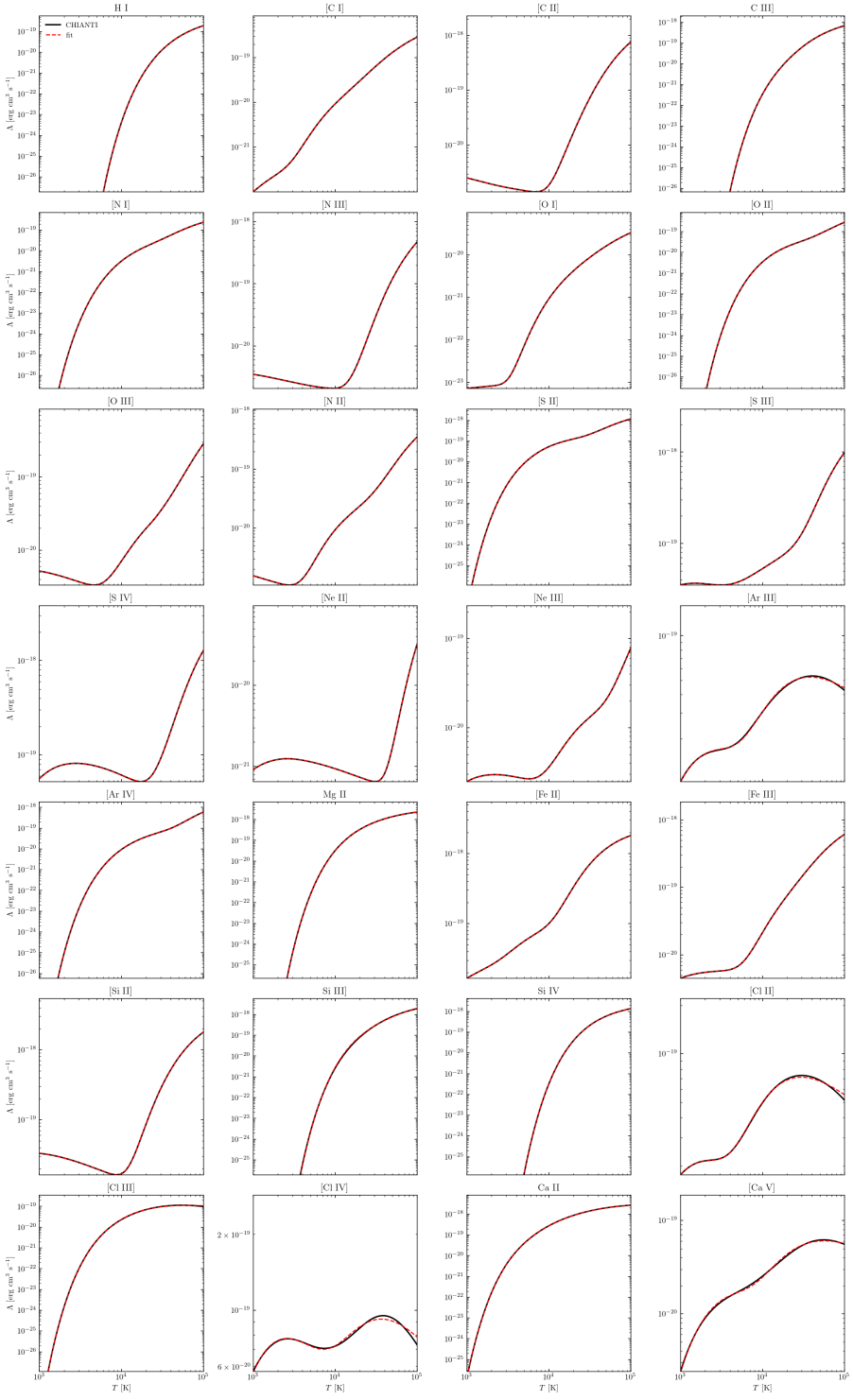


Figure 1: Cooling fits (red dashed) over the direct CHIANTI curves (black), 10^3–10^5 K, in the $n_e \to 0$ limit of statistical equilibrium.

error.

For C I, C II and O I a separate file `data/atomic/excitation/<model>/hcoll_<el>_<stage>.txt` (`make_hcoll_fine_structure.py`) adds atomic-hydrogen impact rates on the ground-term fine structure; `nlevel_load` reads it when it exists. Atomic-hydrogen impact is included only when `par%neutral_hydrogen_excitation` is enabled. The historical expanded iron files and `fe_levels_full` input selector have been removed; the selected excitation model now determines the iron atom.

4.1 Burgess–Tully scaling and Chebyshev fits

CHIANTI stores Υ on the Burgess–Tully (1992) scaled temperature $s \in [0,1]$. With $e_t \equiv kT/\Delta E$, the forward map is

$$s = \begin{cases} 1 - \dfrac{\ln C}{\ln(e_t + C)}, & \text{types 1, 4,} \\ \dfrac{e_t}{e_t + C}, & \text{types 2, 3, 5, 6,} \end{cases} \tag{4}$$

and the descaling back to Υ is

$$\Upsilon = y \ln(e_t + e)\ [1]; \quad y\ [2]; \quad \frac{y}{e_t + 1}\ [3]; \quad y \ln(e_t + C)\ [4]; \quad \frac{y}{e_t}\ [5]; \quad 10^{y}\ [6], \tag{5}$$

where $y(s)$ is the scaled value. Instead of the spline-through-table evaluation used by CHIANTI itself, the fitting scripts fit $y(s)$ with a Chebyshev series over the image of $T \in [10^3, 10^5]$ K intersected with the tabulated range,

$$u = \frac{2(s - s_{\rm lo})}{s_{\rm hi} - s_{\rm lo}} - 1, \qquad y = \sum_{k=0}^{n_c - 1} c_k\, T_k(u), \tag{6}$$

clipping s into $[s_{\rm lo}, s_{\rm hi}]$ outside the fit domain. The degree grows from 2 until the descaled $\Upsilon(T)$ matches the direct spline evaluation to 0.2% (cap at 12). Weak intercombination transitions whose tables span decades (e.g. O III $^1D_2, ^1S_0 \rightarrow {}^5S_2$) defeat a linear-space fit; for those the series stores $\log_{10} y$ instead, marked by the log flag (`logf=1`, apply 10^y before descaling). The Fortran `nlevel_mod` evaluates exactly this recipe (Clenshaw summation of the series); `verify_nlevel_pyneb.py` replicates it in Python.

The two entries above 1% have benign origins: the O III outlier is the physically negligible $^1S_0$–$^5S_2$ intercombination transition ($\Upsilon \sim 10^{-4}$), and the S II errors reflect the coarse 10-point `.scups` tables of that ion — the deviation is from the interpolating spline, at the level of the resolution of the source data itself.

4.2 Statistical equilibrium and emissivities

Given (T_e, n_e), the solver builds the rate matrix from the file contents alone (plus the atomic-hydrogen rates where an `hcoll` file exists, entering as $n_{\rm HI} q^{\rm H}$ beside $n_e q$): collisional rates from Eq. (1) with de-excitation by detailed balance, $q_{ul} = 8.629 \times 10^{-6}\, \Upsilon / (g_u \sqrt{T})$ and $q_{lu} = (g_u/g_l)\, q_{ul}\, e^{-\Delta E/kT}$, plus the radiative A_{ul}; the level populations follow from $\dot{n}_i = 0$ with the closure $\sum_i n_i = 1$, and the emissivity of a line is $j_{ul} \propto n_u A_{ul}\, \Delta E_{ul}$.

Table 2: *n*-level exports, as recorded in the headers of `data/atomic/nlevel_<el>_<stage>.txt` (generated 2026-07-26). "worst ϒ" is the largest maximum relative error among the fitted transitions of that ion, measured on the descaled $\Upsilon(T)$ against the direct spline evaluation over 10^3–10^5 K. The expanded Fe models reach 0.20% (Fe II, 16 levels) and 0.93% (Fe III, 34 levels).

Ion	levels	A	ϒ fits	worst ϒ	Ion	levels	A	ϒ fits	worst ϒ
H I	25	75	45	0.20%	S IV	5	6	10	0.18%
C I	5	8	10	0.52%	Ar III	5	9	10	0.19%
C II	5	7	10	0.56%	Ar IV	5	10	10	0.20%
C III	5	4	10	0.35%	Mg II	3	2	3	0.15%
N I	5	9	10	0.78%	Fe II	13	40	78	0.20%
N II	5	8	10	0.17%	Fe III	14	38	91	0.20%
N III	5	6	10	0.45%	Si II	5	6	10	0.19%
O I	5	9	10	0.17%	Si III	5	4	10	0.65%
O II	5	9	10	0.27%	Si IV	3	2	3	0.54%
O III	6	10	15	1.68%	Cl II	5	9	10	0.17%
Ne II	2	1	1	0.16%	Cl III	5	10	10	0.19%
Ne III	5	9	10	0.19%	Cl IV	5	9	10	0.19%
S II	5	9	10	1.25%	Ca II	5	7	10	0.19%
S III	6	12	15	0.29%	Ca V	5	9	10	0.19%

5. Verification

5.1 Internal consistency: fitted files vs. direct CHIANTI

Running the same statistical-equilibrium solver twice — once reading the fitted `nlevel` files, once evaluating the raw CHIANTI splines and `.wgfa` rates directly — isolates the error contributed by the fits and the file format. Over the grid $T_e \in \{5, 10, 20\} \times 10^3$ K, $n_e \in \{10^2, 10^3, 10^4\}$ cm^{-3} (measured 2026-07-25, on the `nlevel` files before their regeneration of 2026-07-26; the script that made this comparison is not among `tools/fitting/`, so it was not repeated on the current files):

Diagnostic ratio	worst $\lvert$fit/raw $-1\rvert$
[O II] $\lambda3726/\lambda3729$	0.05%
[S II] $\lambda6717/\lambda6731$	0.52%

The fitted files were faithful to CHIANTI v11.0.2 at that date.

5.2 Cross-check against PyNeb

PyNeb (v1.1.31) carries its own atomic data — for these ions the collision strengths SSB14 (O III), T11 (N II), Kal09 (O II), TZ10 (S II) with A-values FFT04-SZ00, FFT04, Z82-WFD96, RGJ19 respectively — so this comparison is a cross-calibration against an independent reference, not a fit-residual test. Tables 3 and 4 give the classic temperature and density diagnostics, rerun on the current files with `verify_nlevel_pyneb.py` on 2026-09-27: the [O III], [N II] and

[S II] columns reproduce the 2026-07-25 values exactly, and the [O II] column is the rerun (it moved by up to 0.9% with the 2026-07-26 regeneration of `nlevel_o_2.txt`).

The temperature diagnostics agree to 1.2% ([O III]) and 0.3% ([N II]). The density diagnostics drift to 6–10% at $n_e = 10^4$ cm^{-3}, near and above the critical densities of the ^2D levels, where the ratios become A-value-limited — and the two codes take their A-values from different compilations (CHIANTI v11 vs. Z82-WFD96 and RGJ19). Since the fitted files reproduce raw CHIANTI to $\leq 0.5\%$ (previous subsection), this spread measures the state of the atomic data, not the fitting procedure; it should be kept in mind as the floor of attainable agreement when the validation gates compare MoCHII line ratios with PyNeb-based references.

The same script now also covers seven further ratios, with worst differences over the same (T_e, n_e) grid (rerun 2026-09-27): [S IV] 10.51 μm/1406 9.1%, [Ar III] 7136/7751 0.7%, [Ar IV] 4740/4711 4.0%, [Fe III] 4658/4702 16.0%, Si III] 1892/1883 6.9%, [Cl III] 5518/5538 6.6%, [Cl II] 8579/9124 1.5%. The [Fe III] offset (MoCHII 9–16% below PyNeb's BB14 data at every point) has not been traced to a particular A-value or collision strength.

6. Adding an ion

The procedure is a data operation:

1. add one entry to the `IONS` dictionary of `fit_cooling.py` (element, stage; the starting ladder comes from the level energies) and of `fit_nlevel.py` (element, stage, number of levels);
2. run both scripts; check the reported maximum fit errors and the overlay figure;
3. extend `verify_nlevel_pyneb.py` with the diagnostic ratios of the new ion where PyNeb has data, and record the comparison.

A further stage of a registered element, within the stages its `element_<el>.txt` carries, needs nothing more: at setup `species_mod` loads `cooling_<el>_<stage>.txt` for every such stage where it exists, and `lines_mod` solves every stage that has an `nlevel` file. A new element is not only data: it also needs its `data/atomic/element_<el>.txt` (`make_element_data.py`), an `abund_<El>` parameter in `define.f90`, and an entry in the element list of `species_mod`. C, Ar and Fe, which this memo named as the next additions, have been added, and so have Mg, Si, Cl and Ca; the registry is the eleven elements C, N, O, Ne, S, Ar, Mg, Fe, Si, Cl, Ca.

Table 3: Temperature diagnostics vs. PyNeb.

		[O III] $\lambda5007/\lambda4363$			[N II] $\lambda6584/\lambda5755$		
T_e [K]	n_e [cm^{-3}]	MoCHII	PyNeb	diff	MoCHII	PyNeb	diff
5000	10^2	4104	4153	−1.2%	829.6	828.6	+0.1%
5000	10^3	4093	4133	−1.0%	818.3	817.2	+0.1%
5000	10^4	3907	3943	−0.9%	624.7	623.9	+0.1%
10000	10^2	152.0	153.4	−0.9%	66.97	66.93	+0.1%
10000	10^3	151.7	152.8	−0.7%	65.75	65.64	+0.2%
10000	10^4	146.0	147.0	−0.6%	52.23	52.12	+0.2%
15000	10^2	50.97	51.30	−0.6%	29.06	29.08	−0.1%
15000	10^3	50.77	51.10	−0.6%	28.53	28.59	−0.2%
15000	10^4	49.08	49.43	−0.7%	23.26	23.32	−0.3%
20000	10^2	29.82	29.78	+0.2%	19.23	19.23	+0.0%
20000	10^3	29.58	29.67	−0.3%	18.92	18.92	−0.0%
20000	10^4	28.68	28.81	−0.5%	15.71	15.71	+0.0%

Table 4: Density diagnostics vs. PyNeb.

		[O II] $\lambda3726/\lambda3729$			[S II] $\lambda6717/\lambda6731$		
T_e [K]	n_e [cm^{-3}]	MoCHII	PyNeb	diff	MoCHII	PyNeb	diff
5000	10^2	0.7979	0.7641	+4.4%	1.298	1.333	−2.6%
5000	10^3	1.430	1.383	+3.4%	0.8223	0.8396	−2.1%
5000	10^4	2.362	2.491	−5.2%	0.5577	0.5052	+10.4%
10000	10^2	0.7634	0.7411	+3.0%	1.309	1.351	−3.1%
10000	10^3	1.254	1.241	+1.1%	0.8810	0.9139	−3.6%
10000	10^4	2.265	2.397	−5.5%	0.5655	0.5203	+8.7%
15000	10^2	0.7502	0.7341	+2.2%	1.320	1.349	−2.1%
15000	10^3	1.175	1.179	−0.3%	0.9185	0.9520	−3.5%
15000	10^4	2.206	2.345	−5.9%	0.5704	0.5294	+7.7%
20000	10^2	0.7439	0.7319	+1.6%	1.343	1.347	−0.3%
20000	10^3	1.130	1.144	−1.2%	0.9543	0.9786	−2.5%
20000	10^4	2.168	2.311	−6.2%	0.5766	0.5373	+7.3%